BASE CHANGE OF A NORMALIZED DUALIZING COMPLEX ALONG A FLAT LOCAL MAP WITH THE SAME MAXIMAL IDEAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-dualizing/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The base change statement for internal homs along flat maps. Invoked inside the counted proof (source lines 4266-4272 and 4277-4281) as a cross-chapter pointer. Transfers the internal hom computations along the base change.
- The criterion for finite injective dimension of a complex over a Noetherian local ring with a single nonvanishing local cohomology module. Invoked inside the counted proof (source lines 4273-4276) as a cross-chapter pointer. Gives finite injective dimension of the base change.

Source reference. Original reference more-algebra-lemma-base-change-RHom: sources/stacks-dualizing/pinned-original-source.tex, source lines 2719–2719. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-base-change-RHom, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference more-algebra-lemma-finite-injective-dimension-Noetherian-local: sources/stacks-dualizing/pinned-original-source.tex, source lines 4074–4074. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-finite-injective-dimension-Noetherian-local, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:dualizing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $A \to B$ be a local homomorphism of Noetherian local rings. Let $\omega_A^\bullet$ be a normalized dualizing complex. If $A \to B$ is flat and $\mathfrak{m}_A B = \mathfrak{m}_B$, then $\omega_A^\bullet \otimes_A B$ is a normalized dualizing complex for B.*

Proof. It is clear that $\omega_A^\bullet \otimes_A B$ is in $D^b_{Coh}(B)$. Let κ_A and κ_B be the residue fields of A and B. By More on Algebra, Lemma more-algebra-lemma-base-change-RHom we see that

$$R\mathrm{Hom}_B(\kappa_B, \omega_A^\bullet \otimes_A B) = R\mathrm{Hom}_A(\kappa_A, \omega_A^\bullet) \otimes_A B = \kappa_A[0] \otimes_A B = \kappa_B[0]$$

Thus $\omega_A^\bullet \otimes_A B$ has finite injective dimension by More on Algebra, Lemma more-algebra-lemma-finite-injective-dimension-Noetherian-local. Finally, we can use the same arguments to see that

$$R\mathrm{Hom}_B(\omega_A^\bullet \otimes_A B, \omega_A^\bullet \otimes_A B) = R\mathrm{Hom}_A(\omega_A^\bullet, \omega_A^\bullet) \otimes_A B = A \otimes_A B = B$$

as desired. □